Outputs

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Regression

- Classification:
  - Split to minimise Gini impurity or maximise information gain
  - Leaf gives answer as most common class to reach it

Regression

- Classification:
  - Split to minimise Gini impurity or maximise information gain
  - Leaf gives answer as most common class to reach it
- Regression:
  - Split to minimise variance or maximise information gain
  - Leaf gives answer as mean value to reach it

    (median may confer an advantage — any idea when?)
- Otherwise identical!

Variance reduction

- Variance of output measures how consistent a node is...

  ...so choose splits that minimise it

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  similarly for right, σ_r^2 (Y_l = data that goes left)

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- Minimise weighted combination:

$$L(\texttt{split}) = \frac{n_l}{n}\sigma_l^2 + \frac{n_r}{n}\sigma_r^2$$

  $n =$ total exemplar count
  $n_l =$ exemplars traveling down left branch
  $n_r =$ exemplars traveling down right branch

Information gain

- Information gain is not often used for regression tasks with decision trees, as it is not immediately applicable

[1] https://www.biopsychology.org/norwich/isp/chap8.pdf

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In order to apply,

- First fit Gaussian distribution to output variable
- Then, compute entropy [1]

$$\frac{1}{2}\log\left(2\pi e\sigma^2\right)$$

- Information gain, thus, is

$$I(\texttt{split}) = \frac{1}{2}\log\left(2\pi e\sigma_p^2\right) - \frac{n_l}{2n}\log\left(2\pi e\sigma_l^2\right) - \frac{n_r}{2n}\log\left(2\pi e\sigma_r^2\right)$$

(same variables as previous slide, with p subscript for parent)

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From Decision Trees to Random Forests via Bias-variance tradeoff

University of Bath

Bias-Variance Tradeoff

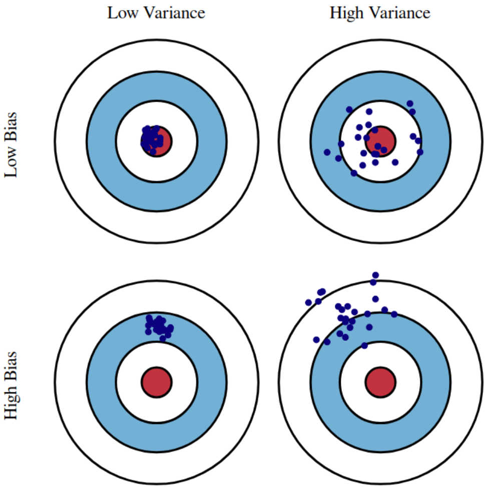

Figure: Pictorial depiction of the components of bias-variance tradeoff